DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Supplementary Material

A. Cross-dataset generalization

We evaluate zero-shot transfer of Waymo-trained checkpoints on 150 nuScenes validation scenes. Each scene supplies one fixed, continuous 20-frame clip from FRONT, FRONT_LEFT, and FRONT_RIGHT. Native camera timestamps give a clip duration of approximately 1.67 s. Inputs at $\{0, 5, 10, 15\}$ leave 16 targets: 150 clips contain 1,800 context and 7,200 target images, including 5,400 interpolation and 1,800 extrapolation targets. Outputs are 240×160.

The supplied camera and actor states and shared state adapters match the Waymo conditions. All networks, codecs, and normalization parameters remain frozen, without target-domain training or scene adaptation. Table S1 reports means. Ours improves over DGGT by 1.42 dB PSNR and 0.014 SSIM. Each dataset retains its native camera timing.

Table S1. Zero-shot reconstruction on nuScenes: 150 scenes and 7,200 target images.

Method	PSNR ↑	SSIM ↑
STORM	18.14	0.672
DGGT	24.91	0.791
StreetForward	23.19	0.715
Ours	**26.33**	**0.805**

B. Training details

Learning is divided into four stages so that the completion target has a stable meaning. Recovery determines the evidence and support, the codec defines its latent coordinates, and the completion network learns a distribution over the missing codes. The last stage adapts this distribution to real observations without changing the representation it predicts.

Data and splits. Counts in Table S2 include only data passing quality checks. The original Waymo training set is divided into 740 training and 58 development sequences; the 58 development sequences select checkpoints, and the 202 validation sequences are used for testing. Synthetic maps, base vehicle models, and human identities are separated by source asset. Texture variants of a vehicle and clothing/motion variants of a human remain in one split. Real logs are deduplicated and separated by road segment, continuous capture, and identifiable actors. Completeness refers to geometry, materials, and motion bindings within the finite query support.

The synthetic test split supplies 100 scenes for paired hidden-surface evaluation and 20 scenes for asset editing. Fixed-asset reuse is also evaluated on 20 real scenes (Appendices H and I).

Table S2. Data and supervision. Scene maps and independent assets are split by source identity.

Data	Train	Dev.	Test	Supervision
Waymo sequences	740	58	202	RGB, calibration, ego poses, LiDAR, vehicle boxes and tracks
Synthetic scenes	10,000	1,000	1,000	Geometry, materials, depth, semantics, instances and actor states
Synthetic maps	80	10	10	Static background geometry and road structure
Vehicle assets	10,000	1,000	1,000	Canonical geometry, materials, dimensions and rigid frames
Human identities	2,500	250	250	Shape, texture, template, skeleton, skinning and joint states

Complete targets on a fixed layout. Partial input establishes $\mathcal{Q}$. Both branches use the same query positions, scales, types, order, component ownership, and human bindings, following the same block order and within-block indices as the partial input. Complete meshes are queried on this fixed support. For each non-sky query, the owning component's mesh is expressed in the query frame and intersected with its cell B_i. Positive-area triangle intersections supply surface targets. A cell with no intersecting surface receives a surface-free target after asset completeness checks.

For an intersecting triangle fragment f of area A_f, we draw

$$n_f = \max\left(1, \left\lceil \frac{64 A_f}{\delta_{\min}^2} \right\rceil\right) \tag{S1}$$

deterministic area samples, where $\delta_{\min}$ is the smallest cell side. Asset, query, and triangle indices determine sample locations. We render RGB, camera-axis depth, instance identity, and triangle identity using original materials and

fixed illumination. Eight azimuths and three elevations form the base views. Local isolated views cover remaining fragments, providing features for surfaces occluded in the base views.

The Stage 1 visual encoder, frozen after recovery training, maps rendered RGB to 128-dimensional features. At each surface sample, we extract these features and append its normalized depth residual relative to the query center, clipped to $[-1, 1]$. We first average valid views of each sample with observation weights, then aggregate samples by fragment area to obtain $\mathbf{h}_i^{\text{full}} \in \mathbb{R}^{129}$. The partial branch retains query-center projection and visibility-weighted aggregation.

Surface-free cells use a 129-dimensional zero feature and explicit zero-opacity supervision. Surface-free targets describe the absence of mesh surface; free-space labels are supplied by observed rays. Sky targets come from the complete environment map with zero depth residual. Both branches share the visual feature basis, input format, and type-specific codec. The target code is

$$\mathbf{z}_{1,i} = E_{k_i}([\mathbf{h}_i^{\text{full}}; \mathbf{q}_i; \boldsymbol{\delta}_i]) \in \mathbb{R}^{128}. \tag{S2}$$

The 135-dimensional input passes through a $135 \to 128 \to 128$ MLP and LayerNorm. Biases allow a zero feature to encode a nonzero latent. Hard projection separately enforces zero free-space codes during completion.

Table S3. Network and support configuration. Observed anchors and canonical human-template queries augment the base grids.

Module	Configuration
Visual recovery	Width 128; patch size 16; 4 heads; 1 image-attention layer and 2 cross-view/time-summary layers
Identity and human state	24 identity slots; 10-dimensional shape code; 17-joint interface
Query features and latents	Feature dimension 129; latent dimension 128; type-specific $135 \to 128 \to 128$ encoders with LayerNorm
Completion network	Width 128; 4 layers; 4 heads; 16 learnable memory tokens
Gaussian decoding	4 children per query; 23 outputs per child, including 12 appearance parameters
Directional appearance	Constant RGB and three local directional coefficients per channel
Base supports	Background $20 \times 12 \times 6$ ($5 \times 5 \times 3$ m cells); rigid $8 \times 4 \times 4$ inside each asset box; sky 4×4 per face
Sampling	24 explicit Euler steps with observed-code restoration and free-space clearing after each step

Stage 1: visual recovery. Real training logs and synthetic scenes supervise the image encoder, geometry heads, identity slots, and state heads. Real depth comes from training LiDAR, and vehicle states come from annotations. Confidence-weighted real semantic and human-joint auxiliary labels complement synthetic shape, skeleton, binding, and joint supervision. One trajectory-level matching assigns slots to identities and is reused across timestamps. Confidence is trained or calibrated against depth, occlusion, and reprojection reliability. Table S4 lists the loss weights.

Table S4. Stage 1 supervision weights.

Loss	Weight	Loss	Weight
Depth	1.0	Root rotation	0.2
Normals	0.2	Dimensions	0.1
Semantics	0.1	Human shape	0.1
Instances	0.1	Joints	0.2
Root translation	0.5	Confidence	0.1

Stage 2: latent representation fitting. With recovery frozen, we train the codec on synthetic scenes and complete assets, together with real RGB, depth, and silhouette observations. The objective is

$$\begin{aligned}\mathcal{L}_2 = \mathcal{L}_{\text{rgb}} + 0.1\mathcal{L}_{\text{depth}} + 0.2\mathcal{L}_{\text{silhouette}} \\ + 0.05\mathcal{L}_{\text{empty}} + 0.001\mathcal{L}_{\text{scale}}.\end{aligned} \tag{S3}$$

For rendered RGB $\widehat{\mathbf{I}}$ and depth $\widehat{d}$, the observation losses are masked L1 means,

$$\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \tag{S4}$$

$$\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \tag{S5}$$

Here $\langle \ell \rangle_\Omega$ averages valid elements. Surface-free cells receive squared-opacity loss $\mathcal{L}_{\text{empty}}$, alongside silhouette supervision and Gaussian-scale regularization. At the end of Stage 2, we freeze recovery and the codec and encode fixed complete targets $\mathbf{z}_1$ for prior learning.

Stage 3: completion-prior learning. The GCM learns the unknown-query flow-matching objective in Eq. (9) of the main paper. Every unknown query in an accepted synthetic sample has valid geometry, appearance, and binding targets. Samples with an invalid unknown target are excluded from this stage; the loss selector s_i operates within valid targets. Complete scenes provide cross-component context,

and complete assets supply object geometry and appearance. Conditioning masks simulate contiguous unseen surfaces, background holes, long occlusions, short tracks, and uneven camera coverage. Pose, instance, and confidence perturbations expose the generator to recovery errors.

Stage 4: adaptation through rendering. We differentiate through flow integration, decoding, and rendering to update only the GCM. Complete synthetic samples and real logs from the 740 training sequences are mixed at 1:3. The objective is

$$\mathcal{L}_4 = \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{rgb}} + 0.1\mathcal{L}_{\text{depth}} + 0.05\mathcal{L}_{\text{free}} + 0.1\mathcal{L}_{\text{ground}} + 0.1\mathcal{L}_{\text{contact}} + 0.1\mathcal{L}_{\text{penetration}}. \quad \text{(S6)}$$

The flow term uses complete synthetic samples; each observation or geometry term uses samples carrying its required supervision. Every term is averaged over its own valid elements. With accumulated free-ray opacity a_r, foot signed distance d_f positive above ground, contacting feet Ω_c, and penetration depth d_p positive inside another actor, the geometry terms are

$$\mathcal{L}_{\text{free}} = \langle a_r^2 \rangle, \qquad \mathcal{L}_{\text{ground}} = \langle [-d_f]_+^2 \rangle,$$
$$\mathcal{L}_{\text{contact}} = \langle |d_f| \rangle_{\Omega_c}, \qquad \mathcal{L}_{\text{penetration}} = \langle [d_p]_+^2 \rangle, \quad \text{(S7)}$$

where $[x]_+ = \max(x, 0)$. Frozen decoding and hard code restoration preserve the observed anchors.

Table S5. Four-stage training schedule. R: recovery; C: codec; G: GCM. D_R: real training logs; D_S: synthetic scenes; D_A: complete assets. Batch sizes are global.

Stage	Updated	Frozen	Data	Updates	Batch	Initial LR
Visual recovery	R	–	D_R, D_S	86,208	32	2×10^{-4}
Latent fitting	C	R	D_S, D_A, D_R	57,472	32	3×10^{-4}
Completion prior	G	R, C	Complete D_S, D_A	114,944	32	10^{-4}
Rendering adaptation	G	R, C	$D_S : D_R = 1 : 3$	28,736	8	3×10^{-5}

Optimization and inference. Table S5 gives the four-stage schedule. Training uses eight A100 80 GB GPUs, FP16, and AdamW with $\beta = (0.9, 0.95)$ and weight decay 0.01. Each stage uses warmup followed by cosine learning-rate decay. The 58 development sequences select checkpoints; the 202 validation sequences form the test set. The four stages total 287,360 updates and 1,001.5 GPU hours, including method-specific target preprocessing. Stage costs are 278.6, 184.3, 326.9, and 211.7 GPU hours. Three GCM training seeds share recovery and the codec, giving a combined cost of 2,078.7 GPU hours. The w/o CCM control reuses Stages 1–2 and retrains Stages 3–4 with the same budget. The deterministic w/o GCM control uses the fixed recovery module and codec.

At inference, the log is encoded once. Its query layout and observed codes remain fixed throughout 24 Euler steps, after which type-specific decoders produce assets for repeated state composition.

C. Scene composition details

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. A rigid asset's Gaussian center and covariance transform as

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \quad \text{(S8)}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. Weights satisfy $w_{ij} \geq 0$ and $\sum_j w_{ij} = 1$, giving $\mathbf{A}_i = \mathbf{I}$ in the rest state $\mathbf{P}_j = \mathbf{B}_j$. The transformed center is $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w \mathbf{A}_i \bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$. Its inverse $\mathbf{J}_i^{-1}$ maps viewing directions into the canonical frame; the direction is normalized before appearance evaluation. The inverse is evaluated for nonsingular skinning transforms. The root pose controls overall placement, while the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. External policy and dynamics can advance the state:

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \quad \text{(S9)}$$
$$s_{k+1} = F_{\text{dyn}}(s_k, \pi(o_k), \Delta t). \quad \text{(S10)}$$

Camera and actor state changes reuse the same background and canonical assets. Road meshes and collision proxies use the same coordinates as the visual components, allowing their motion to be driven by a physical simulator.

D. Visual recovery and aggregation details

Input conventions and preprocessing. The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}_{c_v}^r$. Rig poses $\mathbf{T}_r^w(t)$ are supplied in the controlled evaluations relative to the first frame,

giving camera poses $\mathbf{T}^w_{c_v}(t) = \mathbf{T}^w_r(t)\mathbf{T}^r_{c_v}$. Transforms act on column vectors and distances are in meters. Images follow a rectified pinhole model and linear RGB; distortion, rolling shutter, and exposure differences are handled before evidence aggregation.

Shared visual encoding. The Driving Log Encoder uses patch tokenization [2] and transformer attention [4], with image-position, camera-calibration, and time embeddings. Within-image attention extracts spatial features. A second transformer processes pooled image summaries across views and timestamps, and returns the resulting context to the patch features. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. The confidence determines which predictions can contribute surface evidence.

Learned actor queries attend to all image tokens to form trajectory-level identity slots. Each slot then reads the features at an individual timestamp to estimate its state. This separates persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Root poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}^w_r(t)$. A human shape code selects an identity-specific template with fixed canonical support, skeleton topology, and binding. Slot capacity bounds the number of represented actors.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}^w_i$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}^w_i$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Metric support and anchor selection. The background base grid spans $[-20, 80] \times [-30, 30] \times [-3, 15]$ m with $20 \times 12 \times 6$ cells, giving $5 \times 5 \times 3$ m cell scales. Up to 2,048 observed anchors add 0.20 m support cells. Each vehicle uses an $8 \times 4 \times 4$ grid inside its actual bounding box, with cell sides $(L/8, W/4, H/4)$ and up to 512 observed anchors of side 0.20 m. Human queries lie on the canonical template with support neighborhoods of side 0.10 m. Sky has six cubemap faces with 4×4 direction queries per face. Background attention groups use 4 m spatial bins; this grouping scale is independent of the support-cell scales.

Valid estimated depths generate anchors in each branch's coordinates. We deduplicate by the bucket index $\lfloor \mathbf{p}/0.20 \rfloor$, retaining the first point in each bucket. If the unique-bucket count exceeds the branch cap, we select equally spaced indices in the ordered unique-bucket list. Anchor positions retain their sampled coordinates. Human queries carry template-provided binding weights before completion; all four decoded children inherit them. The root pose supplies global placement, and the internal root joint retains its rest transform.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \mathrm{xyz}(\mathbf{T}^w_a(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \mathrm{xyz}(\mathbf{T}^w_h(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{S11}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r^2_{i,t,v}/2), \tag{S12}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{S13}$$

The query confidence is the maximum over accepted observations. The main reconstruction protocol uses center-ray rule C with $\eta = 0.15 + 0.02d$ meters and confidence at least 0.8. A reliable query-center ray votes for free space when

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{S14}$$

Positive observation weight labels a query observed. Among the remaining queries, at least two free-space votes establish a free label; all others remain unknown. Sky projection uses camera rotation, reliable sky semantics, and zero depth residual, and receives no free-space votes.

Footprint rule and downstream reconstruction. We compare center-ray voting with a footprint rule on the same fixed query layouts in 100 synthetic test scenes. The footprint rule places all eight cell vertices using the query's rigid

or skinning transform and computes

$$z^{\max}_{i,t,v} = \max_{x \in B_i(t)} z_v(x), \qquad \Omega_{i,t,v} = \pi_v(B_i(t)). \quad \text{(S15)}$$

The projected footprint is rounded outward to cover every intersected depth pixel. A view votes only when the footprint lies fully inside the image, all depths are valid, every confidence is at least 0.8, and sky and ego-vehicle pixels are absent. It must satisfy

$$z^{\max}_{i,t,v} < \min_{p \in \Omega_{i,t,v}} \left[D_{t,v}(p) - 0.15 - 0.02 D_{t,v}(p) \right]. \quad \text{(S16)}$$

Other views abstain. The two-vote rule, observed-surface priority, and hard clearing are retained.

Table S6. Free-space classification errors on 100 synthetic scenes. Errors count cells containing reference surfaces that are labeled free; categories are mutually exclusive.

Category	Surface-containing cells	Center-ray errors	Footprint errors
Thin structures	18,437	614	9
Occlusion boundaries	27,619	587	7
Other regions	53,128	173	3
Total	99,184	1,374	19
Error / surface-containing cells		1.385%	0.019%
Number of free labels		47,286	32,871
Surface contamination / free labels		2.906%	0.058%

The footprint rule reduces occupied-cell errors from 1,374 to 19 (Table S6). With reference depth, center-ray voting still mislabels 934 occupied cells, localizing an error source to lateral cell coverage. Continuous reference-depth footprint bounds yield zero such errors on this audit set. The remaining 19 errors with discrete estimated depth arise mainly at subpixel structures and depth boundaries.

From mask labels to reconstructed surfaces. The two rules start from the same intermediate state before the first hard clearing. We fix weights, query layout, input depth, postprocessing, and per-query random noise. On 202 Waymo sequences, F improves full-image/dynamic PSNR by 0.03/0.04 dB; on the same 100-scene hidden-region evaluation, it improves hidden PSNR by 0.18 dB, coverage by 1.2 percentage points, and F-score by 1.3 points (Table S7). Thin structures, occlusion boundaries, and human surfaces gain 2.7, 2.2, and 1.0 points of hidden-surface coverage. Human semantic slices can overlap the structural slices and are scored separately. All coverage entries here use a 2 cm point-to-mesh threshold; human asset editing uses 1 cm.

Table S7. **Downstream effects of free-space voting.** Shared checkpoints and per-query noise, with nine seed combinations. Coverage and F-score use 2 cm.

Metric	C	F
Main full-image PSNR ↑	28.63	**28.66**
Main dynamic PSNR ↑	25.17	**25.21**
Construction (s) ↓	**2.66**	2.86
Hidden PSNR ↑	23.18	**23.36**
Hidden LPIPS ↓	0.192	**0.185**
Hidden coverage ↑	95.0%	**96.2%**
Hidden F-score ↑	88.6%	**89.9%**
Thin-structure coverage ↑	92.1%	**94.8%**
Boundary coverage ↑	93.0%	**95.2%**
Human coverage ↑	95.8%	**96.8%**

The mask audit counts 99,184 reference-surface-containing query cells. Its disjoint categories describe cell-label errors; hidden-surface coverage instead measures reference-surface recall using area-uniform samples. The 1,355 net reduction in mislabeled cells (98.62%) and the 1.2-point coverage gain measure these different stages. Ray violations use the 2,020,000 fixed-ray denominator in Appendix J.

On Waymo input views, F gives 30.41 dB PSNR and 121.5 mm depth MAE, versus 30.39 dB and 122 mm for C. Added occlusion changes from 0.18% to 0.17%, and strict free-ray violations from 0.07% to 0.08%. Both rules preserve observed Gaussian parameters exactly. Construction takes 2.86 s with F versus 2.66 s with C, a 7.52% increase on the RTX 4090. Appendix J reports the rendering measurements, and Appendix K specifies timing conditions.

E. Gaussian decoding details

A Gaussian stores its center $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$, opacity α, and directional appearance.

Background, rigid, and human Gaussian decoders independently map each query to $K = 4$ child Gaussians. Each child has 23 output parameters: three center offsets, three scales, four rotation parameters, one opacity, and twelve appearance coefficients. For child k of query i, the center is confined to its support cell:

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{S17}$$

Positive scales and normalized rotations determine positive-definite covariance. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. Human children inherit the skinning weights of their support query; rigid children share their asset's rigid transform. A separate sky head predicts cubemap colors, with the observation-supported domain retained in V_{sky}.

Appearance uses a constant RGB term and three local directional coefficients per channel, evaluated as $c(d) = \max(0, c_0 + C_{\text{app}}\hat{d}_{\text{local}})$. Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

F. Reconstruction protocols and baselines

The main comparison reconstructs scenes under supplied camera and actor states. The last four frames evaluate conditional temporal extrapolation. Table S10 lists test conditions. Target human joints come from independent offline annotation/fitting, and human shape and binding are recovered from context. Test LiDAR is reserved for evaluation. All test-time network weights remain fixed.

Training resources. We retrain STORM, DGGT, and StreetForward from random initialization, preserving their native architectures. These three baselines and Ours use the same real training logs D_R, synthetic scenes D_S, and complete assets D_A. Baselines receive RGB, depth, silhouette, and complete-asset target views through their native objectives. Ours additionally uses cell-empty and template-binding supervision. Checkpoints are selected on the shared development set within each fixed training budget (Table S8). Matching recovery, codec, data, and optimization in the CCM ablation isolates the contribution of component exchange.

Table S8. Training resources for one complete checkpoint, including method-specific target preprocessing. Ours and w/o CCM include the shared recovery and codec stages.

Method	Training objective	Updates	Global batch	GPU hours
STORM	Native reconstruction plus same-source synthetic views	234,816	16	916.7
DGGT-G	Native geometry/rendering plus same-source synthetic views	212,672	16	1,128.4
StreetForward	Native scene reconstruction plus same-source synthetic views	246,144	8	1,036.9
Ours	Four-stage training	287,360	By stage	1,001.5
w/o CCM	Shared R/C; retrained GCM prior and adaptation	287,360	As Ours	977.1

Full generative pipelines. DGGT-G denotes the Gaussian branch of DGGT; DGGT-Full retains its diffusion refinement. STORM+Difix applies DIFIX3D+ [5] to STORM renders. Main Table 2 compares these complete outputs at identical offsets and reference populations. Table S14 gives training/adaptation costs, excluding historical pretraining of frozen foundation models.

Shared state adapter. The baselines use a common deterministic state adapter. Visible Gaussian rendering contributions are projected into the same context instance masks and accumulate weighted identity votes. Vehicles are canonicalized with the inverse supplied root SE(3), transforming centers, covariances, and directional coordinate bases together. Human Gaussians obtain 17-joint skinning weights by barycentric interpolation on corresponding triangles of the context-recovered identity template, followed by inverse skinning. The adapter preserves predicted appearance and adds no prediction network. Unbound outputs remain scored, and actor edits that cannot be assembled count as failures.

For the same Gaussian set, inverse and forward placement produce a maximum linear-RGB difference below 4×10^{-6}. Converting a native time-varying representation into fixed canonical assets changes cross-time reconstruction as shown in Table S9. The adapted column is used in the main comparison.

Table S9. Effect of conversion to fixed canonical assets. PSNR is in dB.

Method	Native	Adapted	Change
STORM	25.94	25.79	-0.15
DGGT-G	27.51	27.35	-0.16
StreetForward	26.02	25.88	-0.14
Ours	–	28.63	–

Table S10. Test conditions shared by DriveCompose, STORM, DGGT, and StreetForward.

Condition	Shared setting
Context RGB	The same four timestamps and three cameras
Camera and ego state	Shared intrinsics, extrinsics, and ego poses
Test depth/LiDAR	Not an input; depth is estimated from RGB
Identity, boxes, root tracks	Shared supplied states
Human shape and binding	Recovered from context; common binding interface
Target human joints	Shared independent offline annotations/fits
Test-scene optimization	None for any method

The first 200 valid synchronized frames per test sequence form ten non-overlapping 20-frame clips, totaling 2,020 clips. Context indices are $\{0, 5, 10, 15\}$ for three cameras. Twelve interpolation frames yield 72,720 images, and the four extrapolation frames $\{16, 17, 18, 19\}$ yield 24,240, totaling 96,960 targets. Every target timestamp has valid dynamic masks and depth-evaluation samples. Group-specific metrics in Table S11 use 3:1 interpolation/extrapolation weights; full-image SSIM is reported separately as the method-level aggregate from main Table 1. Depth scale alignment is used only for evaluation, with no feedback to reconstruction or rendering.

Table S11. Temporal interpolation and conditional extrapolation. Overall full-image SSIM aggregates both time groups; all other metrics are group-specific. Depth is scale-aligned.

Method	Time group	Full PSNR	Overall full SSIM	Dynamic region PSNR/SSIM	Depth AbsRel/δ_1
STORM	Interp.	26.18	0.782	22.86/0.629	0.186/0.747
STORM	Extrap.	24.62		20.94/0.581	0.218/0.691
DGGT	Interp.	27.73	0.875	23.89/0.656	0.227/0.717
DGGT	Extrap.	26.21		22.29/0.604	0.271/0.657
StreetForward	Interp.	26.27	0.819	21.86/0.619	0.205/0.735
StreetForward	Extrap.	24.71		19.78/0.559	0.249/0.679
Ours	Interp.	**28.96**	**0.903**	**25.61/0.875**	**0.089/0.889**
Ours	Extrap.	**27.64**		**23.85/0.843**	**0.105/0.857**

Depth scale and metric evaluation. The aligned depth columns use a positive per-frame scale on a fixed valid LiDAR mask M_t,

$$s_t = \text{median}_{p \in M_t} \frac{d_t(p)}{\widehat{d}_t(p)}. \tag{S18}$$

Only scale is fitted. Table S12 uses the same masks and frame–clip–sequence aggregation for aligned and unaligned depth. Alignment is applied inside the depth evaluator; reconstruction, composition, surface distances, and RGB reprojection use the original metric coordinates.

Table S12. Depth evaluation with and without per-frame scale alignment.

	Per-frame scale aligned		Unaligned	
Method	AbsRel ↓	δ_1 ↑	AbsRel ↓	δ_1 ↑
STORM	0.194	0.733	0.249	0.669
DGGT	0.238	0.702	0.319	0.614
StreetForward	0.216	0.721	0.278	0.657
Ours	**0.093**	**0.881**	**0.119**	**0.847**

Training repeats and shared costs. Table S13 uses the main protocol: 202 sequences, 2,020 clips, and 96,960 targets per group. The 29-timestamp ablation retains its separate protocol in Appendix G. Ours and w/o CCM share 462.9 GPU hours for Stages 1–2. Their three-seed pipelines cost $462.9+3(1001.5-462.9) = 2078.7$ and $462.9+3(977.1-462.9) = 2005.5$ GPU hours, respectively. Deduplicating the shared stages across both gives $2078.7 + 2005.5 - 462.9 = 3621.3$ GPU hours (Table S14).

Table S13. Three training repeats for main-task full-image PSNR. SD is the sample standard deviation of the three training means.

Method	R0	R1	R2	Mean	SD
STORM	25.73	25.75	25.89	25.79	0.09
DGGT	27.32	27.29	27.44	27.35	0.08
StreetForward	25.86	25.81	25.97	25.88	0.08
Ours	28.66	28.54	28.69	28.63	0.08

Table S14. Training/adaptation costs in GPU hours: per-checkpoint mean, training SD, and total for the three-seed pipeline. Historical pretraining of frozen foundation models is excluded.

Method	Per checkpoint	SD	Three-seed total
STORM	916.7	6.41	2750.1
DGGT-G	1128.4	8.39	3385.2
StreetForward	1036.9	9.71	3110.7
DGGT-Full	1202.83	10.02	3608.5
STORM+Difix	978.03	7.84	2934.1
Asset Harvester	821.97	11.12	2465.9
w/o CCM	977.1	11.05	2005.5
Ours	1001.5	9.32	2078.7

State-input sensitivity. Table S15 fixes target views and compares supplied states, context-predicted states, and two perturbation levels. Context prediction has 54.3 mm MPJPE and 4.87° joint-rotation error. Ours drops from 25.17 to 23.83 dB dynamic PSNR (1.34 dB), retaining 1.14/0.85 dB over DGGT/w/o CCM. At the two perturbation levels, its gains over DGGT are 1.29/0.85 dB and over w/o CCM are 0.87/0.72 dB.

Table S15. State-input sensitivity on 202 sequences with identical target views. All entries are dynamic PSNR (dB); σ_p/σ_R denote position/rotation noise SD.

State	STORM	DGGT	StreetForward	w/o CCM	Ours	Δ_{DGGT}	$\Delta_{\text{w/o CCM}}$
Supplied	22.38	23.49	21.34	24.18	25.17	1.68	0.99
Context predicted	21.81	22.69	20.73	22.98	23.83	1.14	0.85
$\sigma_p = 5$ cm, $\sigma_R = 3°$	21.97	22.94	20.87	23.36	24.23	1.29	0.87
$\sigma_p = 10$ cm, $\sigma_R = 6°$	21.18	22.06	20.01	22.19	22.91	0.85	0.72

G. Ablation protocols and seed statistics

The two learned variants use training seeds $\{0, 1, 2\}$ and generation seeds $\{7, 17, 27\}$ for each checkpoint, giving nine combinations. We first average generation seeds, then frames, windows, sequences, and training seeds equally. In Tables S16 and S17, each entry is the mean ± sample SD of the three training-seed means; parentheses give the average sample SD across generation seeds. Zero variation for w/o GCM describes deterministic inference with shared, fixed recovery and codec. The $T = 8$ entries in Table S17 are the corresponding results in Table S16.

Table S16. Component ablations: mean ± training SD (mean generation SD).

Metric	w/o GCM	w/o CCM	Ours
Full-image PSNR	27.137 ± 0	28.286 ± 0.074 (0.032)	**28.913 ± 0.061 (0.028)**
Full-image SSIM	0.867 ± 0	0.893 ± 0.001 (0.0006)	**0.911 ± 0.001 (0.0005)**
Full-image LPIPS	0.183 ± 0	0.147 ± 0.002 (0.0013)	**0.129 ± 0.002 (0.0011)**
Dynamic PSNR	23.431 ± 0	24.683 ± 0.112 (0.048)	**25.617 ± 0.095 (0.041)**
Dynamic SSIM	0.811 ± 0	0.849 ± 0.002 (0.0010)	**0.873 ± 0.002 (0.0008)**

Table S17. Input-timestamp sensitivity with fixed checkpoints: mean ± training SD (mean generation SD).

T	Variant	PSNR ↑	SSIM ↑	LPIPS ↓
2	w/o CCM	25.237 ± 0.103 (0.055)	0.817 ± 0.003 (0.0011)	0.254 ± 0.004 (0.0023)
2	Ours	**26.341 ± 0.089 (0.047)**	**0.851 ± 0.002 (0.0009)**	**0.213 ± 0.003 (0.0019)**
4	w/o CCM	27.126 ± 0.086 (0.041)	0.869 ± 0.002 (0.0008)	0.187 ± 0.003 (0.0018)
4	Ours	**27.864 ± 0.072 (0.035)**	**0.891 ± 0.001 (0.0006)**	**0.159 ± 0.002 (0.0014)**
8	w/o CCM	28.286 ± 0.074 (0.032)	0.893 ± 0.001 (0.0006)	0.147 ± 0.002 (0.0013)
8	Ours	**28.913 ± 0.061 (0.028)**	**0.911 ± 0.001 (0.0005)**	**0.129 ± 0.002 (0.0011)**

Each of the 202 test sequences contributes one 29-timestamp window. Inputs $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$ share seven target timestamps $\{2, 6, \ldots, 26\}$ and three cameras, giving 4,242 target images per combination. Two learned variants, three input settings, and nine seed combinations yield 54 groups and 229,068 predictions. The deterministic w/o GCM control adds 4,242, totaling 233,310.

Non-renderable counts are $0/10,908$ learned-variant window instances and $0/202$ for w/o GCM. Low quality, holes, and contact errors remain scored. All variants share preprocessing, 240×160 resolution, annotation-based dynamic masks, and equal sequence weighting. Training samples $T \in \{2, 4, 8\}$ uniformly, and inference uses 24 Euler steps. Hyperparameters are selected only on the development split.

At $T = 2, 4, 8$, CCM increases full-image PSNR by 1.104, 0.738, and 0.627 dB and reduces LPIPS by 0.041, 0.028, and 0.018 (Table S17). The largest gains occur with the sparsest input.

H. Off-trajectory rendering and completion analysis

Each offset direction renders $2,020 \times 3 \times 20 = 121,200$ images and 6,060 videos; both directions total 242,400 images and 12,120 videos per offset magnitude and group. Videos contain 20 frames at 10 Hz and never concatenate clips. The reference population consists of original-trajectory recordings from the same test clips, cameras, and timestamps. FID uses a fixed Inception-v3 with 2,048-dimensional features; FVD uses a fixed I3D with 1,024-dimensional features. Left and right scores are computed separately and averaged, without pooling the two populations first.

One hundred synthetic test scenes separately provide paired evaluation of regions never observed in the inputs (Table S18), using representative shifts of ±2 m. FID/FVD describe distribution quality, whereas hidden-region metrics evaluate scene-corresponding completion.

Table S18. Paired evaluation of synthetic hidden regions.

Variant	PSNR	LPIPS	Coverage	F-score@2 cm
w/o GCM	16.82	0.401	61.5%	51.7%
w/o CCM	21.56	0.236	89.7%	81.3%
Scene latent	22.24	0.221	91.8%	84.1%
Object coupling	22.51	0.212	92.4%	85.0%
Ours	**23.18**	**0.192**	**95.0%**	**88.6%**

Fixed-reference evaluation. Full generative pipelines and Gaussian branches are compared on one reference-defined region (Table S20), independent of method coverage; missing outputs are retained in its denominator. Ours achieves the lowest RGB MAE (0.0213) and LPIPS (0.144), 0.0023 and 0.019 below STORM+Difix. The common-valid-support ablation in Table S19 is a separate diagnostic.

Table S19. Linear-RGB reprojection error and coverage. The common support covers 58.7% of the reference set.

Variant	Common MAE	Own MAE	Coverage
w/o GCM	0.022	0.024	62.1%
w/o CCM	0.028	0.041	86.9%
Ours	**0.019**	0.029	**92.4%**

Table S20. Scene correspondence on a fixed reference region: 202 sequences and bidirectional 2 m shifts.

Method	RGB MAE ↓	LPIPS ↓
STORM	0.0354	0.232
DGGT-G	0.0306	0.208
StreetForward	0.0369	0.246
DGGT-Full	0.0242	0.171
STORM+Difix	0.0236	0.163
Ours	**0.0213**	**0.144**

Predicted surfaces and geometry metrics. We construct surfaces by rendering and fusion. Gaussians with opacity at least 0.01 are evaluated within 3σ. Six axis-aligned orthographic views are rendered in blocks over a common evaluation domain, with pixel size at most 1 mm inside each block. A ray's first accumulated-opacity crossing of 0.5 defines a surface depth; rays below this opacity have no predicted surface. We fuse depths into a sparse TSDF with 2 mm voxels and an 8 mm truncation distance, and extract the mesh in valid, supported voxels, leaving unknown regions open. Native neural fields use the same accumulated-opacity threshold, fusion, and distance evaluation.

Predicted and reference meshes each supply 65,536 area-uniform samples. A BVH computes each sample's nearest distance to the other triangle mesh. Coordinates are in meters and distances are non-squared Euclidean distances. Evaluation keeps the original metric scale and alignment. With predicted samples P, reference samples G, and corresponding surfaces S_{pred} and S_{GT} in the evaluation region,

$$P_\tau = \frac{1}{|P|} \sum_{p \in P} \mathbf{1}[d(p, S_{\text{GT}}) < \tau], \tag{S19}$$

$$R_\tau = \frac{1}{|G|} \sum_{g \in G} \mathbf{1}[d(g, S_{\text{pred}}) < \tau], \tag{S20}$$

$$F_\tau = \frac{2P_\tau R_\tau}{P_\tau + R_\tau}. \tag{S21}$$

Coverage is reference-surface recall R_τ. We compute metrics per scene and seed before averaging. Asset coverage uses only surfaces assigned to that actor, with 2 cm vehicle and 1 cm human thresholds. Complete reference geometry and input visibility define newly visible surfaces; reference surfaces outside query support remain in the coverage denominator.

Real-view correspondences. Independent reference geometry establishes correspondences between recorded and off-trajectory views. Dynamic actors are transformed with shared states. The reference evaluator removes occlusion-ambiguous correspondences for every method; finite outputs and rendering coverage determine method validity. Test LiDAR is used only by the evaluator. Linear-RGB MAE is measured on both each method's valid subset and their intersection, which covers 58.7% of the reference set (Table S19). CCM reduces common-support MAE from 0.028 to 0.019 and raises overall coverage from 86.9% to 92.4% relative to w/o CCM.

Task-adapted completion mechanisms. The scene-latent control adapts the scene-level Gaussian latent generation mechanism of CompleteSplat [3]; the object-coupling control adapts JRM's cross-object latent coupling [6]. Both operate through the shared recovery module, codec, query layout, and state interface. They replace the completion interaction mechanism while using the same input evidence, training data, supervision, and 24-step sampling setup. The shared recovery and codec stay fixed. The completion training schedule is given in Appendix B, and comparison scores average three training and three generation seeds.

Table S18 uses the same 100 synthetic test scenes and 12,000 shifted target images per seed combination. Each learned variant produces 108,000 outputs across nine seed combinations; scene-wise scores include all views, frames, and seeds. Over the object-coupling control, CCM adds 0.67 dB PSNR, reduces LPIPS by 0.020, and increases coverage and F-score by 2.6 and 3.6 percentage points. The PSNR gain over the scene-latent control is 0.94 dB. Surface evaluation uses original metric coordinates, 65,536 area-uniform samples from each mesh, and a shared 2 cm point-to-triangle threshold.

Visibility-stratified synthetic evaluation. Table S21 reuses the same 100 synthetic test scenes, ± 2 m shifts, inputs, checkpoints, and generated samples as the paired hidden-region evaluation. Each scene supplies one 20-frame clip with three cameras and context timestamps $\{0, 5, 10, 15\}$ (12 images). Targets cover both shift directions, all 20 timestamps, and three cameras, giving 12,000 images per seed combination. Context timestamps are also valid targets because the shifted cameras differ from the input cameras. Three training seeds and three generation seeds reuse trained weights without adapting on these scenes.

Visibility v is the fraction of a fixed reference surface region's area covered by the union of input views. Well-observed regions have coverage $\geq 80\%$, severely missing regions $\leq 20\%$, and intermediate regions $20\% < v < 80\%$. Reference-surface correspondences assign target pixels to background, vehicle, or human and to a visibility group. Methods use the same valid masks; group-wise frame means are averaged with equal scene weight. Group PSNRs and full-image PSNR are computed independently. The hidden-region test requires exactly 0% input coverage, whereas the severely missing group permits up to 20%.

CCM adds 1.02, 1.27, and 1.53 dB for severely missing background, vehicle, and human regions (Table S21). Intermediate regions gain 0.58, 0.72, and 0.84 dB; well-observed regions gain 0.04, 0.05, and 0.03 dB. Benefits increase as input coverage decreases.

Table S21. PSNR stratified by component and input visibility.

Component and visibility	w/o CCM	Ours	Gain
Background, well observed	30.42	30.46	+0.04 dB
Background, intermediate	26.28	26.86	+0.58 dB
Background, severely missing	22.17	23.19	+1.02 dB
Vehicle, well observed	28.56	28.61	+0.05 dB
Vehicle, intermediate	24.77	25.49	+0.72 dB
Vehicle, severely missing	21.29	22.56	+1.27 dB
Human, well observed	27.11	27.14	+0.03 dB
Human, intermediate	23.49	24.33	+0.84 dB
Human, severely missing	19.85	21.38	+1.53 dB

Read-scope interventions on Waymo. Table S22 uses the same 202 Waymo windows as the main $T = 8$ ablation. Each 29-timestamp window uses $\{0, 4, 8, 12, 16, 20, 24, 28\}$ from three cameras (24 inputs), with targets $\{2, 6, 10, 14, 18, 22, 26\}$ (4,242 images per seed combination). Training seeds $\{0, 1, 2\}$ and generation seeds $\{7, 17, 27\}$ give nine combinations. Recovery, codec, query layout, target states, and 24-step sampling are fixed.

Reading only background, other vehicles, or other humans applies inference-time masks to the same Ours checkpoints, without separate retraining. Local attention and the global log condition remain unchanged. The w/o CCM endpoint uses the per-component-memory control retrained with the same budget. The w/o CCM and Ours endpoints reuse the corresponding results in main Table 3. Waymo read interventions and synthetic visibility scores are reported separately; component gains overlap.

Table S22. CCM read-scope diagnostics.

Extra readable components	Full-image PSNR
No exchange (w/o CCM)	28.286
Background	28.721
Other vehicles	28.574
Other humans	28.352
All components	**28.913**

Dense attention and memory capacity. Table S23 fixes queries, sampling steps, and scenes to compare interaction mechanisms. CCM-16 is 0.05 dB below dense global attention, with $1.93\times$ faster completion and 42.71% lower peak memory. This query-controlled test is separate from the full-query results in Table S18. At 350,000 queries, dense attention/CCM take 73,627.4/1,380 ms (Table S24). Increasing memory from 16 to 64 tokens adds 0.06 dB while raising completion time from 1,380 to 2,087.1 ms (Table S25).

Table S23. Completion mechanisms with 16,384 queries and 24 sampling steps on 100 synthetic scenes. Completion time is in ms and peak memory in GB. Independent/Dense denote per-component memory/global attention.

Mechanism	PSNR ↑	LPIPS ↓	Cov. (%) ↑	Time ↓	GB ↓
Independent	20.92	0.257	88.31	253.7	3.79
Scene latent	21.79	0.234	90.43	301.4	4.07
Object coupling	22.02	0.224	91.17	386.8	4.56
Dense	**22.49**	**0.207**	**93.71**	361.3	6.93
CCM-16	22.44	0.209	93.46	**187.2**	3.97

Table S24. Query scaling on 100 fixed inputs. Dense attention and CCM use identical query counts within each row. Times are in ms and peak memory in GB.

Queries	Dense ms	CCM ms	Dense GB	CCM GB	Speedup
8,192	147.4	116.3	4.18	3.36	1.27
16,384	361.3	187.2	6.93	3.97	1.93
32,768	924.7	319.4	9.17	4.73	2.9
65,536	2846.3	537.1	11.61	6.08	5.3
350,000	73627.4	1380	21.36	14.2	53.35

Table S25. Memory capacity with full-query evaluation on 100 synthetic scenes. Time is completion latency (ms); memory is peak GB.

Tokens	PSNR ↑	LPIPS ↓	Cov. (%) ↑	ms	GB
8	22.95	0.202	94.63	1247.3	13.87
16	23.18	0.192	95	1380	14.2
32	23.23	0.189	95.31	1623.4	14.81
64	23.24	0.188	95.43	2087.1	15.94

I. State-driven asset reuse

Paired synthetic edits. Twenty identity-disjoint synthetic test scenes each contain one vehicle and one human. A 1 m camera shift, 30° vehicle rotation, and human pose change reuse fixed completed assets. Joint completion leads all six newly visible surface metrics (Table S26), improving over Asset Harvester by 0.14/0.35 dB vehicle/human PSNR and 0.37/0.89 percentage points coverage.

Table S26. Newly visible asset surfaces after editing. Twenty identity-disjoint synthetic scenes; vehicle/human coverage thresholds are 2/1 cm.

Method	Veh. PSNR ↑	LPIPS ↓	Cov. ↑	Human PSNR ↑	LPIPS ↓	Cov. ↑
STORM	18.24	0.353	66.74%	17.86	0.377	62.83%
DGGT	19.71	0.307	74.23%	18.95	0.331	69.47%
StreetForward	18.53	0.339	68.17%	18.15	0.363	65.29%
w/o GCM	17.8	0.362	63.4%	17.12	0.391	60.8%
w/o CCM	22.46	0.218	91.8%	21.38	0.247	88.6%
Asset Harvester	24.04	0.181	95.73%	22.71	0.215	93.41%
Ours	**24.18**	**0.176**	**96.1%**	**23.06**	**0.201**	**94.3%**

Twenty real and twenty identity-disjoint synthetic test scenes each contain one tested vehicle and human. We evaluate the original state, a 1 m lateral camera shift, a 30° vehicle rotation about its root and the world vertical axis, and a human pose change at fixed root placement. Every operation reuses the same completed asset and generation seed under fixed lighting, with appearance held fixed. Complete reference geometry defines newly visible surfaces as described in Appendix H.

Real-scene edit ratings. Training seed 0 and generation seed 7 give forty edits (Table S27). Two evaluators blinded to method identity independently inspect predefined target views; a third adjudicates disagreements. Structural integrity fails when a continuous hole, detached surface, or joint tear spans at least five pixels at the evaluation resolution. Ours succeeds in all twenty vehicle rotations and seventeen human edits. Its three human failures involve a large elbow bend, leg self-occlusion, and loose-clothing boundaries. Vehicle dimensions and human shape parameters remain fixed, and previously observed human texture has canonical linear-RGB MAE 0.014.

Table S27. Structural integrity of real actor edits with training seed 0 and generation seed 7.

Variant	Vehicle	Human	Overall
w/o GCM	14/20	10/20	60.0%
w/o CCM	18/20	15/20	82.5%
Ours	**20/20**	**17/20**	**92.5%**

Seed variation and scene clustering. The three training seeds $\{0, 1, 2\}$ and three generation seeds $\{7, 17, 27\}$ yield 360 edits on the same twenty real scenes (Table S28). Vehicles succeed in 177/180 edits (98.33%), humans in 157/180 (87.22%), and the aggregate in 334/360 (92.78%). The sample SD of the three training-seed mean structural-integrity rates is 0.48 percentage points. Scene-cluster bootstrap sampling retains all eighteen edits for a selected scene, giving a 95% structural-integrity interval of 85.28–99.17%. The 26 failures are concentrated in four scenes with 9, 8, 6, and 3 failures.

Table S28. Structural-integrity successes across training and generation seeds. Each row evaluates twenty vehicles and twenty humans.

Training	Generation	Vehicle	Human	Total
0	7	20	17	37/40
0	17	20	18	38/40
0	27	19	17	36/40
1	7	20	18	38/40
1	17	19	17	36/40
1	27	20	18	38/40
2	7	20	17	37/40
2	17	20	18	38/40
2	27	19	17	36/40
All combinations		177/180	157/180	334/360

Newly visible texture, identity, and geometry. An asset surface is newly visible when it is invisible in every input observation and visible after editing. Canonical surface correspondences and visibility tests define these regions. Five independent raters inspect newly visible texture, appearance identity, and local geometry; each dimension passes with at least four positive judgments. Texture ratings inspect blur, repetition, stretching, and seams. Identity ratings inspect vehicle shape and paint, and human clothing, body shape, and visible identity features. Geometry ratings inspect wheel/body connections, limb volume, joint bends, and intersections.

Table S29 scores 360 actor edits from twenty scenes and nine seed combinations; camera shifts add no actor-edit trials. Joint acceptance is the per-edit intersection of the four criteria, yielding 298 passes (82.78%). The 62 failures comprise 26 structural failures and 36 structurally intact edits that fail at least one other quality check.

Table S29. **Real actor-edit quality under rule C.** Twenty scenes, nine training/generation-seed combinations, and 360 actor edits. Joint acceptance is the per-edit intersection.

Criterion	Vehicle (180 edits)	Human (180 edits)	All (360 edits)
Structural integrity	177 (98.33%)	157 (87.22%)	334 (92.78%)
New-surface texture	166 (92.22%)	154 (85.56%)	320 (88.89%)
Appearance identity	174 (96.67%)	170 (94.44%)	344 (95.56%)
Local geometry	164 (91.11%)	152 (84.44%)	316 (87.78%)
All four criteria	156 (86.67%)	142 (78.89%)	298 (82.78%)

External asset baselines and rater agreement. Table S30 compares baselines adapted to the asset-edit interface, including Asset Harvester [1]. Deterministic baselines average three trained versions, yielding $20 \times 2 \times 3 = 120$ edits; generative versions use three training and three generation seeds, yielding 360. Ours reaches 82.78% joint acceptance, 2.50 percentage points above Asset Harvester, which leads identity preservation by 0.83 points. Table S31 reports training SD for joint acceptance separately from structural-integrity SD above. Table S32 covers $3 \times 120 + 3 \times 360 = 1440$ outputs across six methods; the 20 scenes are the independent sampling units, with seeds and outputs treated as within-scene repetitions.

Table S30. Real asset edits on 20 scenes, each with one vehicle and one human. Cells give passes/trials (rate). Deterministic/generative methods have 120/360 edits.

Method	Structure	New texture	Identity	Geometry	All four
STORM	77/120 (64.17%)	74/120 (61.67%)	106/120 (88.33%)	81/120 (67.5%)	53/120 (44.17%)
DGGT	89/120 (74.17%)	84/120 (70%)	111/120 (92.5%)	90/120 (75%)	63/120 (52.5%)
StreetForward	82/120 (68.33%)	77/120 (64.17%)	108/120 (90%)	83/120 (69.17%)	56/120 (46.67%)
w/o CCM	304/360 (84.44%)	293/360 (81.39%)	336/360 (93.33%)	295/360 (81.94%)	259/360 (71.94%)
Asset Harvester	331/360 (91.94%)	317/360 (88.06%)	**347/360 (96.39%)**	314/360 (87.22%)	289/360 (80.28%)
Ours	**334/360 (92.78%)**	**320/360 (88.89%)**	344/360 (95.56%)	**316/360 (87.78%)**	**298/360 (82.78%)**

Table S31. Training SD of basic-protocol joint acceptance, in percentage points.

Method	Training SD (pp)
STORM	5.2
DGGT	2.5
StreetForward	5.2
w/o CCM	1.27
Asset Harvester	2.93
Ours	1.27

Table S32. Rater agreement over 1,440 outputs from six methods on 20 scenes.

Criterion/coefficient	Coefficient	Repeat mean	Repeat SD
Structure, Cohen κ	0.864	0.864	0.001
Texture, Fleiss κ	0.754	0.754	0.005
Identity, Fleiss κ	0.604	0.604	0.011
Geometry, Fleiss κ	0.75	0.75	0.001

Extended state and view protocol. The 360-edit quality evaluation uses fixed target-state and view lists. The expanded protocol applies vehicle yaw changes of 30°, 60°, and 90°, and human maximum non-root joint rotation differences of 15°, 30°, and 45° at fixed root placement. Twenty scenes and nine seed combinations yield $20 \times 9 \times (3+3) = 1,080$ actor edits. Three predefined target views per edit give 3,240 images. Each seed reuses one completed asset across all states, with the five-pixel criterion defined at 240×160 resolution.

Deterministic baselines have $20 \times 3 = 60$ edits per level and 360 overall; generative versions have $20 \times 9 = 180$ per level and 1,080 overall. The three predefined target views belong to the same edit. Tables S33 and S34 report level-wise counts and aggregates. Ours passes structure in 943/1,080 edits (87.31%) and all four criteria in 805/1,080 (74.54%), the latter 4.91 percentage points above Asset Harvester.

Table S33. Complete extended-edit counts. Cells give structural/joint passes. Vehicle 30° and human 15° match the basic protocol.

Method	Edits/angle	Veh. 30°	Veh. 60°	Veh. 90°	Human 15°	Human 30°	Human 45°
STORM	60	45/33	37/23	30/15	32/20	27/15	22/10
DGGT	60	50/37	42/27	36/18	39/26	33/21	27/15
StreetForward	60	46/33	39/24	31/15	36/23	31/18	24/12
w/o CCM	180	163/138	150/123	131/97	141/121	126/102	105/78
Asset Harvester	180	173/155	162/139	151/120	158/134	142/114	120/90
Ours	180	177/156	171/147	159/132	157/142	150/126	129/102

Table S34. Aggregate over six edit levels. Structural/joint rates are percentages; joint training SD is in percentage points.

Method	Edits	Struct. passes	Struct. rate	Joint passes	Joint rate	Train SD
STORM	360	193	53.61	116	32.22	3.94
DGGT	360	227	63.06	144	40	3.63
StreetForward	360	207	57.5	125	34.72	5.02
w/o CCM	1,080	816	75.56	659	61.02	1.53
Asset Harvester	1,080	906	83.89	752	69.63	1.31
Ours	1,080	**943**	**87.31**	**805**	**74.54**	1.37

J. Observation consistency

We fix original cameras, actor states, and evaluation pixels, using the observed-only scene as the pre-completion reference (Table S35). With latent projection, observed Gaussian parameters are preserved exactly, while added Gaussians can change rendering through occlusion and alpha compositing. We fix 10,000 rays per sequence, totaling 2,020,000. Added occlusion means added content produces accumulated opacity above 0.01 more than 5 cm in front of an observed surface. Strict free-ray violation requires opacity above 0.05 more than 10 cm in front of the surface.

For Ours, 0.18% added occlusion corresponds to 3,636 rays and 0.07% strict violation to 1,414. Relative to the pre-completion scene, input-view PSNR decreases by 0.03 dB and depth MAE increases by 2 mm. Removing the free-ray loss raises the two ray rates to 1.34% and 0.91%, showing its effect on rendering beyond query-level parameter preservation. Without latent projection, parameter change is 0.0127, strict violation 0.48%, and depth MAE 132.7 mm.

Table S35. Input-observation preservation under completion and constraint ablations. C/F and the two C ablations use 2,020,000 fixed rays; – denotes unreported measurements.

Configuration	Parameter Δ	PSNR	RGB MAE	Depth MAE (mm)	Occlusion	Violation
Observed only	0	30.42	0.0187	120	0	0
w/o CCM	0	30.368	0.0190	123	0.24%	0.09%
C	0	30.39	0.0189	122	0.18%	0.07%
F	0	30.41	–	121.5	0.17%	0.08%
C, w/o free-ray loss	0	29.97	0.0208	134	1.34%	0.91%
C, w/o latent projection	0.0127	30.09	–	132.7	0.91%	0.48%

K. Efficiency and reuse cost

Training uses 8× A100 80 GB GPUs. Inference timing uses one RTX 4090 24 GB, FP16, batch 1, and 240×160 output, with an average of 1.4 million non-transparent Gaussians per scene. After 20 warmups, 100 repetitions include CPU–GPU transfers and synchronization but exclude disk reads and initial model loading. Construction takes 2.66 ± 0.11 s, state composition plus one camera render takes 7.8 ± 0.4 ms (approximately 128.2 FPS), and peak memory is 14.2 GB.

Let $Q = Q_b + \sum_a Q_a^{\text{vehicle}} + \sum_h Q_h^{\text{human}}$ count non-sky queries. Decoding produces $4Q$ candidate Gaussians, which bound the non-transparent output count. Sky uses the separate cubemap head.

Table S36 breaks down construction. The w/o CCM control takes 2.53 s, so CCM adds 0.13 s (approximately 5.1%). Table S37 computes total time including one construction as $2.66 + 0.0078N$. Here N counts single-camera outputs; a

three-camera query counts as three renders.

Changing only the free-space rule to F raises construction time from 2.66 to 2.86 s (7.52%) under the same hardware, input, and sampling conditions. The main timing breakdown and amortized costs use C.

Table S36. Scene-construction timing breakdown.

Stage	Time
Image preprocessing	0.18 s
Visual recovery	0.64 s
Queries and canonical aggregation	0.17 s
24-step completion	1.38 s
Gaussian decoding	0.26 s
Initial scene registration	0.03 s
Total	**2.66 s**

Table S37. Amortized query costs for fixed completed assets.

N	Total time	Per query
1	2.6678 s	2.6678 s
10	2.738 s	273.8 ms
100	3.440 s	34.4 ms
1,000	10.460 s	10.46 ms

Resolution and external cost comparison. Table S38 separates higher-resolution rendering of the same assets built from 240×160 inputs from reconstruction using 480×320 inputs. At fixed 480 output, higher-resolution input adds 0.37/0.47 dB full-image/dynamic PSNR while increasing construction from 2.66 to 3.92 s. At 480 input/output (Table S39), Ours exceeds DGGT-Full by 1.49 dB full-image and 1.87 dB dynamic PSNR, rendering one image in 13.71 ms versus 161.87 ms.

Table S38. Resolution and cost. 240/480/960 denote 240×160/480×320/960×640. Quality uses 202 sequences; costs use 100 fixed clips on an RTX 4090, FP16, batch 1.

Input/output	Full PSNR ↑	SSIM ↑	Dyn. PSNR ↑	Build s ↓	Render ms ↓	Peak GB ↓
240 to 240	28.63	0.903	25.17	2.66	7.8	14.2
240 to 480, reuse	28.08	0.895	24.56	2.66	13.26	15.07
480 to 480, rebuild	28.45	0.904	25.03	3.92	13.71	18.43
240 to 960, reuse	27.56	0.885	24.01	2.66	29.17	17.61

Table S39. External comparison at 480×320 input and output, with the same hardware, precision, and batch size as the resolution-cost test.

Method	Full PSNR ↑	SSIM ↑	Dyn. PSNR ↑	Build s ↓	Render ms ↓	Peak GB ↓
STORM	25.14	0.769	21.75	1.84	11.63	11.79
DGGT-G	26.71	0.859	22.91	2.16	12.31	13.47
StreetForward	25.21	0.807	20.69	1.95	11.94	12.68
DGGT-Full	26.96	0.869	23.16	2.16	161.87	17.53
Ours	**28.45**	**0.904**	**25.03**	3.92	13.71	18.43

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